PIYUSH GARG

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EDUCATION

Bachelor of Technology in AI & Data Science **Expected Graduation: May 2028**

Akhilesh Das Gupta Institute of Professional Studies

- Relevant Coursework: Deep Learning, Data Structures & Algorithms, Calculus, Probability
- Current CGPA: 7.8/10.0

Central Board of Secondary Education (CBSE 12th) **Graduated: June 2024**

Bal Bhavan International School

- Score: 75%

PROJECTS

Diffusion-Based Generative Modeling for Sparse Detector Data

• Developed a denoising diffusion probabilistic model (DDPM) from scratch using PyTorch.
• Built a generative pipeline including forward diffusion, reverse sampling, and timestep-conditioned denoising.
• Evaluated model performance by comparing statistical properties of generated detector events with real data using energy distribution analysis.

DIV2K Image Super Resolution using SRGAN

• Trained a Super Resolution GAN on the DIV2K dataset for high-quality image reconstruction.
• Used curriculum training strategy (L1 pretraining → adversarial training) for stable GAN convergence.
• Focused on perceptual quality improvement and color fidelity in reconstructed images.

ECG Anomaly Detection using Autoencoders

• Built an autoencoder-based anomaly detection system for ECG time-series signals.
• Used reconstruction error to identify abnormal cardiac patterns in an unsupervised setting.
• Implemented preprocessing pipeline for robust signal normalization and feature extraction.

Fashion-MNIST Image Generation using DCGAN

• Developed a Deep Convolutional GAN to generate synthetic fashion images.
• Designed convolutional generator and discriminator networks.
• Evaluated training stability and sample quality across epochs.

SKILLS

- **Machine Learning:** PyTorch, TensorFlow, Scikit-Learn, Generative Models (GANs, Diffusion, Autoencoders)
- **Programming:** Python, C++, C, JavaScript
- **Scientific Computing:** NumPy, Pandas, Matplotlib
- **Web:** HTML, CSS, React
- **Languages:** English, Hindi